

Chapter 01: An Introduction to relation databases and SQL

Q1. What is role of the database server on client/server database system?

- A. PCs, MACs or workstations on the system
- B. Applications installed on workstations that uses database information
- C. cabling, communication lines, and other components that connect all workstations
- D. None of the above

Answer: B

Q2. What is the role of DBMS in client/server system?

- A. Hosts custom application that uses data from databases
- B. Manages the databases that are stored on the server
- C. Provides API to access data on the server
- D. All of the above

Answer: B

Q3. In a client/server system, the application software is run on _____.

- A. Client
- B. Server
- C. Both client and server
- D. Enterprise server

Answer: A

Q4. The application software on the client uses data that's stored on the database using _____.

Which one best fits the blank space?

- A. a data access API
- B. SQL queries
- C. a database management system
- D. Network

Answer: A

Q5. SQL stands for _____.

- A. Standard Query Language
- B. Special Query Language
- C. Structured Query Language
- D. Structured Query Learning

Answer: C

Q6. Which one is true about SQL?

- A. A programming language allows to build data-driven applications
- B. A language allows any application to communicate with any DBMS
- C. A standard language that is understood by any standard application
- D. All of the above

Q7. Which one is referred to back-end processing in a client/server database system implementation?

- A. The processing that's done by the DBMS
- B. The processing that's done by the client software
- C. The processing that's done by both the DBMS and the client software

D. The processing that is done by neither the DBMS nor the client software

Answer: A

Q8. Which one is referred to as front end in a client/server system?

- A. The client
- B. The server
- C. The communication between the client and the server
- D. None of the above

Answer: A

Q9. In a client/server system, all of the processing is done on the clients.

- A. True
- B. False

Answer: B [In a client/server system, the processing done by an application is typically divided

between the client and the server. Ref.: Page 8]

Q10. Which one is not true about application servers?

- A. provide user interface to users
- B. store business components that do part of the processing of the application
- C. process database requests from the user interface running on the client
- D. If the client request involves accessing data in a database, the application server formulates the appropriate query and passes it on to the database server.

Q11. In a client/server system, which one provides user interface?

- A. The client
- B. The Application Server
- C. The database server
- D. DBMS

Answer: A

Q12. What is ADO.Net?

- A. AN application framework from Microsoft to build distributed applications on top of .Net platform
- B. Microsoft's data access API that can communicate directly with the SQL Server
- C. An API to access remote systems over internet
- D. A component for building web application

Answer: B

Q13. In web-based systems, which one is responsible for sending client request to the web server over internet?

- A. Web service
- B. Network operating system
- C. Web browser
- D. Application server component

Answer: C

Q14. In relational database, data is stored in one or more _____.

- A. Files
- B. Data pages
- C. Tables
- D. Records

Answer: C

Q15. What does a column in a relational table represent?

- A. A collection of entries

- B. One instance of an entity
- C. An attribute of an entity
- D. A single value

Answer: C

Q16. What does a single row in a table represent?

- A. A collection of entries
- E. One instance of an entity
- F. An attribute of an entity
- G. A single value

Answer: B

Q17. What is a primary key?

- A. One or more columns that establishes link between tables
- B. One or more columns that uniquely identify each row in a table
- C. A constraint that enforces a value to be present in a column
- D. A column whose value is automatically generated by the system

Answer: B

Q18. Non-primary key column that does not contain duplicates is defined as _____.

- A. Index
- B. Foreign key
- C. Nullability
- D. Unique key

Answer: D

Q20. What is an index?

- A. It uniquely identifies each row in a table
- B. It enforces enterprise rules on the value of a column
- C. It provides an efficient way of accessing rows in an table based on one or more columns
- D. It establishes relationship between tables

Answer: C

Q21. An index is automatically created for a tables primary key.

- A. True
- B. False

Q22. Relationships exist between _____.

- A. Between the primary keys in two tables
- B. Between the foreign keys in two tables
- C. Between the primary key in one table and the foreign key in another table
- D. All of the above

Q23. If a table has a one-to-one relationship with another table, the data in two tables can be stored in one table.

- A. True
- B. False

Answer: A [Ref. : Page 30]

Q24. Many-to-many relationships can be implemented in SQL Server.

- A. True
- B. False

Answer: B

[no RDBMS can implement many-to-many relationship]

Q25. How is many-to-many implemented?

- A. By breaking down into two one-to-one relationships
- B. By breaking down into two one-to-many relationships
- C. By breaking down into two many-to-many relationships
- D. By using check constraint

Answer: C

Q26. A table can have only one foreign key.

- A. True
- B. False

Answer: B

[Ref. : Page 31, a table can have more than one foreign key column]

Q27. How can you define what type of data a column can store?

- A. By setting its data type
- B. By defining default value
- C. By allowing .null value
- D. By defining check rule

Answer: A

Q28. What is an identity column in SQL Server?

- A. A column that uniquely identifies each row
- B. A column that can store null value
- C. A column that has auto-incremented numeric value and its value is generated by SQL server
- D. A column that has a default value

Answer: D

Q28. What data type only allows value 1 or 0 that represents a True or False value?

- A. Boolean
- B. Tinyint
- C. Bit
- D. Int

Answer: C

Q29. The type and size of the information that can be stored in the column is determined by _____.

- A. Primary key
- B. Default value
- C. Data type
- D. NULL or NOT NULL

Answer: C

Q30. _____ of column defines that value is used if a value is not provided when a row is added to the table.

- A. Identity
- B. Unique key
- C. Data type
- D. Default value

Answer: D

Q31. SQL fall into two categories. They are _____.

- A. data manipulation language (DML)
- B. data definition language (DDL)
- C. XQuery
- D. Object Query

Answer: A, B

Q32. SQL programmers typically work with the DML statements, while database administrators (DBAs) use the DDL statements.

- A. True
- B. False

Answer: B

[Reverse is true, SQL programmers typically work with the DML statements, while database administrators (DBAs) use the DDL statements. Re.: Page 23]

Q33. Which one is NOT a DDL statement?

- A. CREATE statement
- B. ALTER statement
- C. DELETE statement
- D. DROP statement

Answer: C

Q34. Which one is NOT a DML statement?

- A. SELECT statement
- B. INSERT statement
- C. ALTER statement
- D. DELETE statement

Answer: C

Q35. Which statement do you use to delete a database object?

- A. ALTER statement
- B. DELETE statement
- C. DROP statement
- D. TRUNCATE statement

Answer: C

Q36. To create a new row in a table you use _____ statement?

- A. CREATE
- B. INSERT
- C. ADD
- D. ALTER

Answer: B

Q37. To modify data in a table you use _____ statement?

- A. ALTER
- B. MODIFY
- C. UPDATE
- D. INSERT

Answer: C

Q38. The operation which combines the data from the two tables is joined together into a single result set is called _____.

- A. Union
- B. Join
- C. Data set
- D. Merging

Answer: B

Q39. The most common type of join is _____.

- A. Inner join
- B. Outer join

- C. Cross join
- D. Self-join

Answer: A

Q40. _____ returns rows from both tables only if their related columns match.

- A. An inner join
- B. An outer join
- C. A cross join
- D. A full outer join

Answer: A

Q41. _____ returns rows from one table in the join even if the other table doesn't contain a matching row.

- A. An inner join
- B. An outer join
- C. A cross join
- D. A self-join

Answer: B

Q42. Which one is often referred to as query?

- A. SELECT statement
- B. INSERT statement
- C. UPDATE statement
- D. DELETE statement

Answer: A

[Ref.: Page 27]

Q43. The execution of an INSERT, UPDATE, or DELETE statement is often referred to as _____.

- A. a query
- B. an action query
- C. a modification query
- D. a change query

Answer: B

[Ref.: Page 31]

Q44. What is a view?

- A. A predefined select query that's stored in a database
- B. A set of statements, not limited to only one select statement, stored as an object
- C. A special type of procedure that can return a value or a table
- D. a special type of procedure that's executed when rows are inserted, updated, or deleted from a table or when the definition of a database is changed

Answer: A

Q45. What is a user-defined function?

- A. A predefined select query that's stored in a database
- B. A set of statements, not limited to only one select statement, stored as an object
- C. A special type of procedure that can return a value or a table
- D. a special type of procedure that's executed when rows are inserted, updated, or deleted from a table or when the definition of a database is changed

Answer: C

Q46. What is a trigger?

- A. A predefined select query that's stored in a database

- B. A set of statements, not limited to only one select statement, stored as an object
- C. A special type of procedure that can return a value or a table
- D. a special type of procedure that's executed when rows are inserted, updated, or deleted from a table or when the definition of a database is changed

Answer: D